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HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING



MACHINE LEARNING - SEMESTER 242

ASSIGNMENT

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Class: CC01

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Team Members and Task Distribution

Team Member	Student's ID	Tasks	Contributions
Dặng Ngọc Phú	2252617	Model Training, Hyperparameter Tuning, Visualization, Report Writing	Trained and tuned machine learning models using GridSearchCV Evaluated model performance Selected the best model Created visualizations
Phùng Gia Minh Khôi	2252381	Data Preprocessing, Feature Engineering	Cleaned and preprocessed the dataset Performed TF-IDF vectorization Handled missing values Removed duplicates
Nguyễn Lê Khải Trọng	2252850	Data Preprocessing, Report Writing	Cleaned and preprocessed the dataset Wrote the project report Documented the workflow
Đoàn Tấn Sang	2252711	Visualization, Report Writing	Created visualizations (e.g., correlation matrix, feature distributions) Wrote the project report Documented the workflow

1 Github link

https://github.com/Chilly-On/ML_Researchers

2 Introduction

TED Talks have revolutionized the way we exchange ideas, offering a platform where innovators, educators, and thought leaders share transformative insights on topics ranging from science and technology to art and social issues. Much like the automotive industry, which is populated by globally recognized brands each distinguished by unique models, manufacturing years, pricing, performance metrics, and design philosophies, TED Talks are characterized by their diverse themes, speaker backgrounds, presentation styles, and cultural contexts.

This project harnesses Natural Language Processing (NLP) techniques to delve into the vast repository of TED Talks transcripts, aiming to extract meaningful patterns and trends that underpin these influential speeches. By examining various attributes—such as the speaker's professional background, the publication year of the talk, sentiment, key topics discussed, and audience engagement metrics—we seek to draw parallels between the evolution of public discourse and the dynamic nature of modern communication.

The study not only explores the linguistic nuances and stylistic elements that make each talk unique but also investigates how these factors collectively contribute to the overall impact of the presentation. In doing so, it provides a framework for understanding how ideas are communicated and received across different segments of society. The insights derived from this analysis are expected to shed light on the

broader cultural and intellectual trends that influence our contemporary world, ultimately enhancing our comprehension of persuasive communication and its role in shaping public opinion.

3 Flowchart for training model

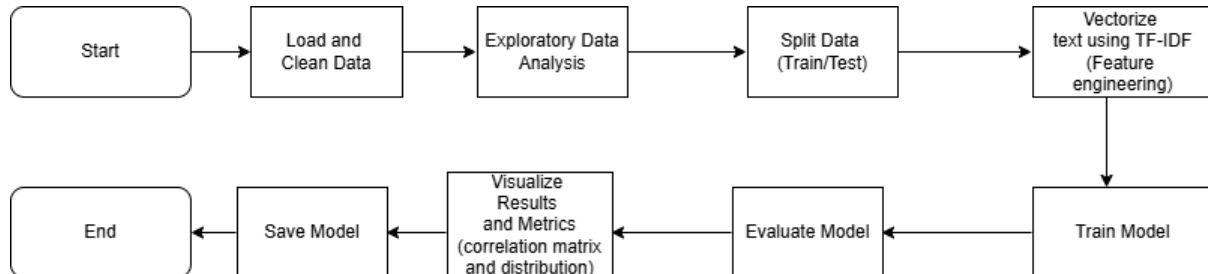


Figure 1: General flowchart for training model

Data Preparation:

- + Collect raw data from TED Talks Transcripts for NLP.
- + Extract the transcription column and remove any missing values from the dataset.
- + Transform the text to lowercase and eliminate any numerical digits and punctuation marks.

Data Preprocessing:

- + Feature Extraction: Use the TfidfVectorizer to turn cleaned text into numbers by calculating TF-IDF scores. It considers up to 5000 features, including both single words (unigrams) and pairs of words (bigrams).
- + Transformation: The vectorizer creates a sparse matrix, which we then convert into a dense array for easier handling.

+ DataFrame Creation:

- *Build a new DataFrame where each column represents a word or n-gram feature.
- *Add a target column (currently set to None) to set the stage for later modeling steps.

Model Evaluation:

It helps in comparing models and fine-tuning hyperparameters during the development phase by visualizing correlation matrix and distribution.

Model Selection To Training :

- + The training involves tuning model parameters through optimization techniques to minimize error or maximize performance.

Final Evaluation:

- + Evaluate the final result after the training to conduct a final evaluation on a hold-out test set that was not used during training or validation. This provides an unbiased estimate of the model's real-world performance before deployment.

4 Code for training model

Clean Data Code & Vectorize text using TF-IDF

```
1 import pandas as pd
```

```
2 from sklearn.feature_extraction.text import TfidfVectorizer
3 import pandas as pd
4 import re
5
6 # Select the transcription column and clean the null in the data
7 train_data = pd.read_csv(f"C:/Users/DELL/Desktop/Machine learning/ML_Researchers-data/
    data/raw/ted_talks_en.csv")
8 df = pd.DataFrame(train_data)
9 print("\nData converted to DataFrame:")
10 df = df[['transcript']]
11 df = df.dropna(subset=['transcript'])
12 print(df.head())
13
14 def clean_text(text):
15     text = text.lower() # Convert to lowercase
16     text = re.sub(r'\d+', '', text) # Remove numbers
17     text = re.sub(r'[\W\s]', '', text) # Remove punctuation
18     text = text.strip()
19     return text
20
21 df['cleaned_transcript'] = df['transcript'].apply(clean_text)
22
23 df = df[['cleaned_transcript']] # Only keep cleaned transcript
24 print(f"Language en:")
25 print(df.head())
26
27 # Split data into features and target variable
28 X = df['cleaned_transcript']
29
30 # Vectorizing text using TF-IDF
31 vectorizer = TfidfVectorizer(max_features=5000, ngram_range=(1, 2)) # Consider unigrams
    and bigrams
32 X_vectorized = vectorizer.fit_transform(X)
33
34 # Convert to dense array
35 X_dense = X_vectorized.toarray()
36
37 # Create DataFrame with words as columns
38 df_tfidf = pd.DataFrame(X_dense, columns=vectorizer.get_feature_names_out())
39 print(X_vectorized.shape)
40 print(df_tfidf.shape)
41 df_tfidf['target'] = None
42
43 # Display a sample
44 print(df_tfidf.head())
```

Model Code

```
1 # Import libraries
2 import sys
3 import os
4 os.chdir(r"C:\Users\Admin\Downloads\ML\assignment\ML_Researchers")
5 import pandas as pd
6 import matplotlib.pyplot as plt
7 import seaborn as sns
8 import joblib
9 import numpy as np
```

```
10 from sklearn.model_selection import train_test_split
11
12 from src.data import load_raw_data, preprocess_data, save_processed_data
13 from src.features import create_features
14 from src.models import train_models, evaluate_models, tune_hyperparameters, save_models
15 from src.visualization import plot_correlation_matrix, plot_feature_distribution
16
17 # Step 1: Load raw data
18 print("Loading raw data...")
19 raw_data = load_raw_data("data/raw/ted_talks_en.csv")
20 print("Raw Data Sample:")
21 print(raw_data.head())
22
23 # Step 2: Define target based on views
24 print("\nDefining target variable...")
25 median_views = raw_data['views'].median()
26 raw_data['target'] = raw_data['views'].apply(lambda x: 1 if x > median_views else 0)
27 print("Target Distribution:")
28 print(raw_data['target'].value_counts())
29
30 # Step 3: Preprocess the data
31 print("\nPreprocessing data...")
32 processed_data = preprocess_data(raw_data, text_column='transcript')
33 print("Processed Data Sample:")
34 print(processed_data.head())
35
36 # Save processed data
37 save_processed_data(processed_data, "data/processed/processed_ted_talks_en.csv")
38 print("\nProcessed data saved to 'data/processed/processed_ted_talks_en.csv'.")
39
40 # Step 4: Feature engineering (TF-IDF vectorization)
41 print("\nPerforming feature engineering...")
42 df_tfidf = create_features(processed_data, text_column='transcript', max_features=5000,
43                             ngram_range=(1, 2))
44 print("TF-IDF Data Sample:")
45 print(df_tfidf.head())
46
47 # Save vectorized data
48 df_tfidf.to_csv("data/processed/vectorized_data.csv", index=False)
49 print("\nVectorized data saved to 'data/processed/vectorized_data.csv'.")
50
51 # Step 5: Split data into train, validation, and test sets
52 print("\nSplitting data into train, validation, and test sets...")
53 X = df_tfidf.drop(columns=['target']) # Features
54 y = df_tfidf['target'] # Target variable
55
56 # Split into train (80%) and test (20%)
57 X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state
58                                                     =42)
59
60 # Split train into train (80%) and validation (20%)
61 X_train, X_val, y_train, y_val = train_test_split(X_train, y_train, test_size=0.2,
62                                                     random_state=42)
63
64 print(f"Train set shape: {X_train.shape}")
65 print(f"Validation set shape: {X_val.shape}")
66 print(f"Test set shape: {X_test.shape}")
67
68 # Step 6: Save test data for submission.ipynb
```

```
66 print("\nSaving test data for submission.ipynb...")
67 test_data = pd.concat([X_test, y_test], axis=1)
68 test_data.to_csv("data/processed/test_data.csv", index=False)
69 print("Test data saved to 'data/processed/test_data.csv'.")
70
71 # Step 7: Hyperparameter tuning using GridSearchCV
72 print("\nPerforming hyperparameter tuning...")
73 best_models = tune_hyperparameters(X_train, y_train)
74
75 # Step 8: Evaluate the best models on the validation set
76 print("\nEvaluating best models on the validation set...")
77 val_results = evaluate_models(best_models, X_val, y_val)
78 print("Validation Set Evaluation Results:")
79 for model, metrics in val_results.items():
80     print(f"{model}: {metrics}")
81
82 # Step 9: Train the best model on the entire train set (train + validation)
83 print("\nTraining the best model on the entire train set...")
84 best_model_name = max(val_results, key=lambda k: val_results[k]['accuracy'])
85 best_model = best_models[best_model_name]
86
87 # Combine train and validation sets
88 X_train_full = pd.concat([X_train, X_val])
89 y_train_full = pd.concat([y_train, y_val])
90
91 # Train the best model
92 best_model.fit(X_train_full, y_train_full)
93
94 # Step 10: Evaluate the best model on the test set
95 print("\nEvaluating the best model on the test set...")
96 test_results = evaluate_models({best_model_name: best_model}, X_test, y_test)
97 print(f"\n{best_model_name} Test Set Evaluation Results:")
98 for model, metrics in test_results.items():
99     print(f"{model}: {metrics}")
100
101 # Step 11: Save the best model
102 print("\nSaving the best model...")
103 save_models({best_model_name: best_model})
104 print(f"Best model ({best_model_name}) saved to 'models/trained/'.")
```